

Joel Eliason, PhD

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Relevant Employment History

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| July 2025 – Present | <p>■ Postdoctoral Research Fellow <i>Johns Hopkins University</i></p> <p>Hybrid mechanistic + ML modeling for pancreatic cancer immunotherapy: a multiscale ODE-based virtual-patient platform (MATLAB/SimBiology) calibrated against a recent neoadjuvant trial (GVAX + nivolumab + low-dose cyclophosphamide), with end-to-end Python infrastructure for inference and simulation. Built qsp-inference, a two-stage Bayesian calibration framework that runs NumPyro/NUTS MCMC over sparse parameter subgraphs and feeds the resulting posteriors into neural posterior estimation (SBI/SNPE) against the ODE simulator. Built MAPLE for LLM-driven parameter extraction from the literature, and qsp-hpc-tools for SLURM-based simulation campaigns with content-hashed caching. Co-developed qsp-codegen (SBML to C++ CVODE; 25–87× speedup over the SimBiology baseline).</p> |
| January 2022 – July 2025 | <p>■ Bioinformatics Graduate Research Assistant <i>University of Michigan</i></p> <p>Methodological focus on Bayesian multilevel models for spatial structure in multiplexed tissue imaging — log-Gaussian Cox processes, multi-subject point-pattern models, and a novel directional asymmetry framework for cell–cell interactions, applied across PDAC, melanoma, NSCLC, and CRC cohorts. Authored and contributed to three open-source R packages distributed via GitHub with CI: SHADE (Stan-based multilevel model for directional spatial associations; <i>PLOS Comp. Biol.</i> 2026), bipps (joint Bayesian point-pattern inference across subjects; <i>Ann. Appl. Stat.</i> forthcoming), and DIMPLE (distance-matrix-based interaction modeling; <i>Patterns</i> 2023). Embedded collaboration with pathologists, immunologists, and biostatisticians.</p> |
| May 2024 – August 2024 | <p>■ Quantitative Systems Pharmacology Intern <i>Johnson & Johnson</i></p> <p>Modeled longitudinal biomarker and exposure–response relationships from Phase 1/2 clinical-trial data for T-cell-redirecting bispecifics in lymphoma, translating PK–PD findings into dose-selection and early-development decisions for clinical and translational stakeholders.</p> |
| January 2021 – January 2022 | <p>■ Biostatistics Research Assistant <i>University of Michigan</i></p> <p>Wrote optimization algorithms in Python for precision-matrix estimation in Gaussian graphical models, focused on scaling to high-dimensional regimes.</p> |
| November 2018 – July 2020 | <p>■ Software Engineer <i>Pariveda Solutions</i></p> <p>Built a serverless lead-routing system on AWS with a Node.js/Angular admin UI, processing thousands of sales leads per day. Wrote Python and SQL pipelines that surfaced ML model predictions to business users in production.</p> |

Relevant Employment History (continued)

May 2015 – August 2015

■ **Research Intern** *Applied Invention (project for DuPont Pioneer)*

Wrote a Python regression model that predicted corn growth stages from agronomic and weather covariates, beating the prior production baseline and saving irrigation cost in field deployment.

Education

- July 2025 ■ **Ph.D., Bioinformatics, University of Michigan**
- April 2022 ■ **M.Sc., Biostatistics, University of Michigan**
- June 2017 ■ **M.Sc., Applied Mathematics, Northwestern University**
- April 2016 ■ **B.Sc., Mathematics, Brigham Young University**

Selected Software & Open-Source Projects

■ **qsp-inference** — *Bayesian inference and SBI for mechanistic models*

Two-stage hybrid mechanistic+ML calibration framework: NumPyro/NUTS MCMC over sparse parameter-target subgraphs with hierarchical partial pooling, then Gaussian-copula priors feeding neural posterior estimation (SBI/SNPE) against physics-grounded ODE simulators. Includes simulation-based power analysis and optimal Bayesian experimental design. Python.

■ **qspio-pdac** — *virtual-patient platform for pancreatic cancer immunotherapy*

Multiscale ODE-based virtual-patient platform with PDAC-specific tumor-immune-stroma dynamics (CAF-driven desmoplasia, CXCL12 T cell exclusion, exhausted T cells), calibrated against the JHU neoadjuvant GVAX + nivolumab + cyclophosphamide trial. MATLAB / SimBiology.

■ **qsp-hpc-tools** — *HPC orchestration and caching for large-scale simulation*

SLURM-over-SSH orchestration for high-throughput simulation campaigns with three-tier content-hashed caching and drop-in MATLAB and C++ qsp_sim backends (25–87× speedups). Python.

■ **qsp-codegen** — *SBML to C++ CVODE code generator (co-developed)*

Generates C++ CVODE simulators from SimBiology-exported SBML with sympy/CSE analytical Jacobians, bundled with a model-agnostic qsp_sim_core driver consumed via CMake find_package. Python + C++.

■ **MAPLE** — *Model-Aware Parameterization from Literature Evidence*

Schema-validated LLM pipeline that extracts mechanistic parameters from the literature into structured calibration targets, linking source measurements to forward models with an 8-axis relevance rubric that produces translation sigmas for downstream Bayesian inference. Python; pip + MCP.

■ **R packages** — **spatial Bayesian methods for tissue imaging**

Authored and co-authored R packages for multiplex tissue imaging analysis: **SHADE** (Stan-based multilevel Bayesian model for asymmetric, directional cell–cell spatial associations; *PLOS Comp. Biol.* 2026), **bipps** (Bayesian inference of multivariate point patterns across subjects; *Ann. Appl. Stat.* forthcoming), and **DIMPLE** (distance-matrix-based quantification and visualization of spatial cellular interactions; *Patterns* 2023).

Selected Research Publications

Journal Articles

- 1 **Eliason, J.**, Frankel, T., Rao, A. & Peruzzi, M. Joint Modeling of Spatial Dependencies Across Multiple Subjects in Multiplexed Tissue Imaging. *Annals of Applied Statistics*. Forthcoming (2026).

- 2 **Eliaison, J.**, Peruzzi, M. & Rao, A. SHADE: A Multilevel Bayesian Framework for Modeling Directional Spatial Interactions in Tissue Microenvironments. *PLOS Computational Biology* **22**, e1013930. ISSN: 1553-7358. (2026) (Feb. 2026).
- 3 **Eliaison, J.** & Popel, A. S. Structured Schemas for LLM-Modeler Collaboration in Quantitative Systems Pharmacology Model Calibration. *bioRxiv*. Preprint (2026).
- 4 **Eliaison, J.** *et al.* Characterizing Spatial Immune Architecture in Metastatic Melanoma Using High-Dimensional Multiplex Imaging. *Frontiers in Immunology* **16**, 1560778. ISSN: 1664-3224. (2025) (Apr. 29, 2025).
- 5 **Eliaison, J.** & Rao, A. Investigating Ecological Interactions in the Tumor Microenvironment Using Joint Species Distribution Models for Point Patterns. *The New England Journal of Statistics in Data Science* **2**. Publisher: New England Statistical Society, 296–310. ISSN: 2693-7166 (June 2024).
- 6 Masotti, M., Osher, N., **Eliaison, J.**, Rao, A. & Baladandayuthapani, V. DIMPLE: An R package to quantify, visualize, and model spatial cellular interactions from multiplex imaging with distance matrices. *Patterns* **4**. Publisher: Elsevier. ISSN: 2666-3899 (Dec. 2023).

Conference Proceedings

- 1 Chetty, V., **Eliaison, J.** & Warnick, S. *Passive reconstruction of non-target-specific discrete-time LTI systems in 2016 American Control Conference (ACC)* ISSN: 2378-5861 (July 2016), 66–71.

Skills

Statistical & ML	Bayesian inference, multilevel/hierarchical and longitudinal modeling, GLMMs, regression and regularization, survival/time-to-event analysis, biomarker clustering and phenotyping, spatial statistics, point processes, simulation-based inference, machine learning, deep learning.
Clinical & biomedical data	Cohort-level analyses across multiple disease areas (PDAC, melanoma, NSCLC, CRC); biomarker–outcome modeling from Phase 1/2 clinical-trial data; multiplexed tissue imaging; cross-subject/cross-cohort partial pooling; sensitivity analyses and confounder/bias diagnostics.
Hybrid SciML & mechanistic modeling	Mechanistic ODE/PDE modeling combined with data-driven inference; neural posterior estimation (SBI/SNPE) against physics-grounded simulators; Quantitative Systems Pharmacology (QSP), pharmacometrics / PK–PD, virtual-patient simulation; uncertainty quantification.
Methods development & scientific writing	Novel algorithms and models for Bayesian calibration of mechanistic models and spatial cell–cell interaction inference, published in peer-reviewed venues (<i>PLOS Comp. Biol.</i>, <i>Ann. Appl. Stat.</i>, <i>Patterns</i>) and released as open-source tools (SHADE, bipps, DIMPLE, MAPLE); reproducible, well-documented workflows with CI and versioning.
Computational	Python, MATLAB, SimBiology, C++, R, Stan, NumPyro, SBML, SQL, Bash, Git, $\LaTeX$.
Systems	HPC / SLURM orchestration, AWS, large-scale simulation campaigns, software architecture, API development, CI/CD.

Skills (continued)

Languages	■ Fluent in English, proficient in Russian.
Miscellaneous	■ Academic research, teaching, scientific writing, statistical consulting.